Project Closure and Recap Form

***This form should be completed after reaching the final milestone and upon completing your project at the end of the semester*. A single form must be submitted by each team.**

First, please fill the following entries with information about your project:

- **Group #**: 6
- **Group Members**: Stella Allingham, Ava Crowley, Crystal Li, Kenlee Duong, Joann Dao
- **Site URL**: https://termproject-is424.web.app/index.html
- **GitHub Repo URL:** https://github.com/allingham-design/houseflow-termproject

Next, engage in self-reflection by addressing the following questions:

1. ***What worked well, and why?***

Working with the client worked well for us for this project. We had a mutual connection to him, and he was also a UW student, so we found it relatively easy to work with him, since he was our age and had similar life circumstances and environments to many of us. Given that we are all college students with roommates as well, we also understood his business need and problem very well, so that made it easier for us to work on his project, knowing that we too could see the need for the system.

2. ***What didn't work well, and why not?***

Working together didn't work super well for us. We found it sort of difficult to split up the work evenly and even with that, it was challenging to get everyone to complete their work. Often times, we were working at the last minute to finish milestones, and different people were tasked with different things each time, despite our efforts to assign roles and split up the work. I think if we had more in-person group meetings, it would be easier to work together, but when group mates are not physically together, it is difficult to collaborate on projects like this.

3. ***If you were to do the same project again, what would you do differently?***

If we were doing the same project again, we would set group meeting times more consistently. I think this would help with all group members being on the same page, and same timeline. Given the fact that everyone is busy with their own individual schedules, team members weren't in synch or on the same page when working individually. It would be helpful if we sticked to a more consistent meeting timeline and itinerary, which would also help with workflow and sharing responsibilities.

4. ***What are your takeaways after completing the project?***

Our takeaways after completing the project are the importance of teamwork in a collaborative setting. We learned how important is it to rely on your team, delegate work when necessary, and pull your own weight. We also learned the importance of communication and setting clear, and early deadlines. Communicating with both our team members as well as our client was important for our project's success. We also learned how crucial adaptability is in group settings. When conflict arises, we changed direction and continued moving forward.

5. ***Describe a scenario that you thought would be hard to go through, but it turned out to be easier than originally thought. Describe another for the other way around.***

Something we thought would be hard to go through but turned out to be easier than originally thought was working with our client. When we first learned we had to collaborate with an external party for this project, we figured it would be difficult to meet and get clear feedback, but since the client was also a college student with similar experiences and environments as us, it was much easier to get his attention and thoughts regarding our project. Something we thought would be easy to go through but turned out to be harder than originally thought was collaborating with our group. Because we are all in class together, we figured it would be relatively simple to work together on this project, but it quickly became obvious that everyone has different class schedules, extracurriculars, and personal schedules, so it was harder than we thought. While this was a struggle all semester, by the end of the semester and project, we did get into a good routine with working together.

6. ***Have you showed your project to people outside your class? What was the impression?***

We have not yet showed our project to people outside of our class, but we have talked about and described our project to people outside of our class. The impression has been positive, and most of our roommates, family and friends have been supportive and excited to see and hear about our progress. I think it is impressive to a lot of people outside of our class just how much we are able to accomplish using the resources and knowledge we've learned here as students.

7. ***What other features would you like to add if you have more time?***

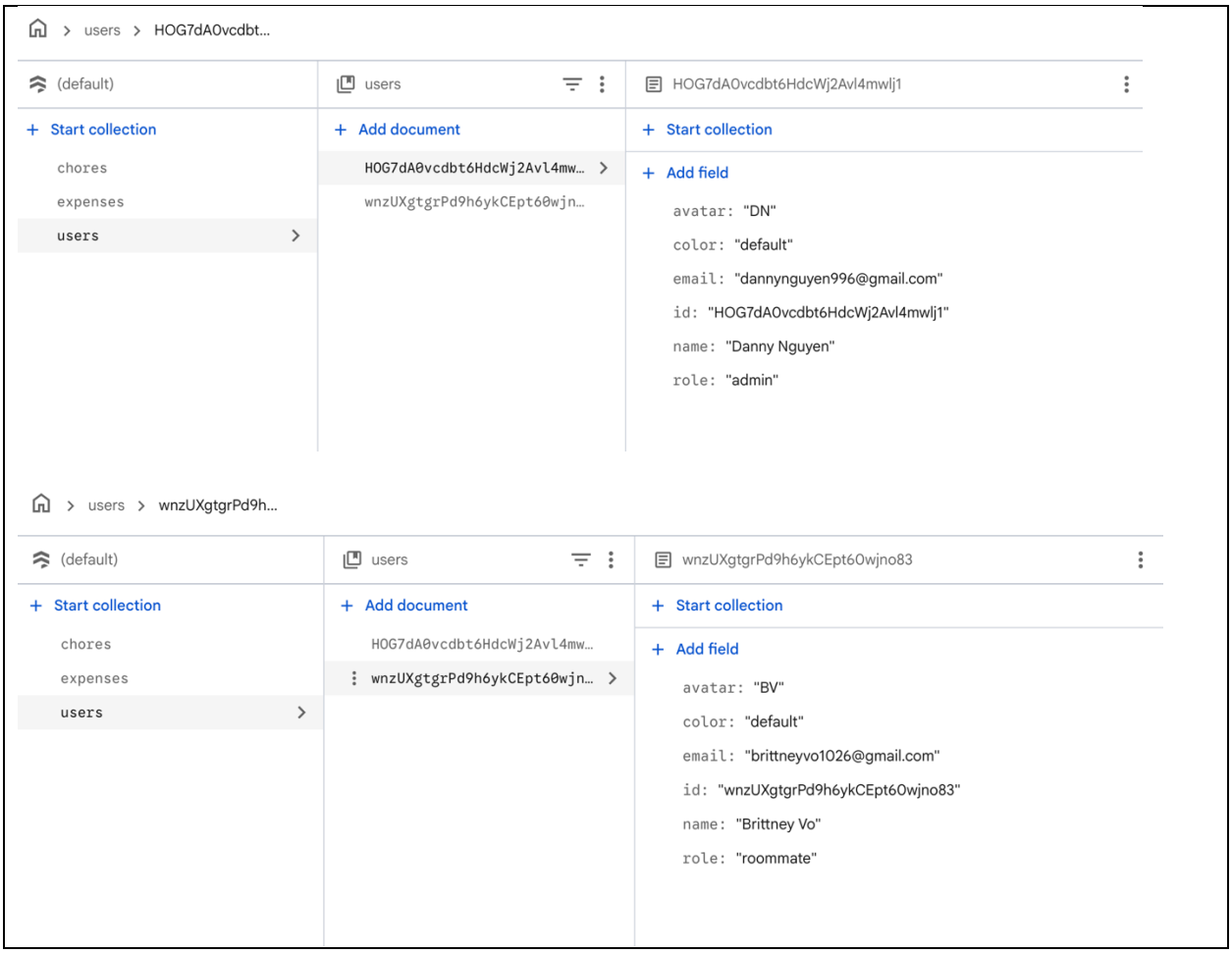
If we had more time to work on our project, I think we would like to add a function that automatically updates the expenses and rent information upon payment. Currently, the site functions where just the admin can edit rent information on the expenses page, and the roommates are assigned their monthly portion of the rent, and instructed to Venmo the admin, using the Venmo QR code on the site. Then, the admin has to go into the site and mark the roommates as complete/paid, once they have Venmo'd him the appropriate amount. This is one of the functionalities that only the admin can do, but ideally, with more time and knowledge, we would like to make this function automatic, where the site automatically updates and marks the roommates "paid" once they have sent their money in. We didn't have time for this, or the knowledge to know how to make this work, so we have decided our method is the best it can be for now, and we are happy with it.

8. ***Anything else you want to say?***

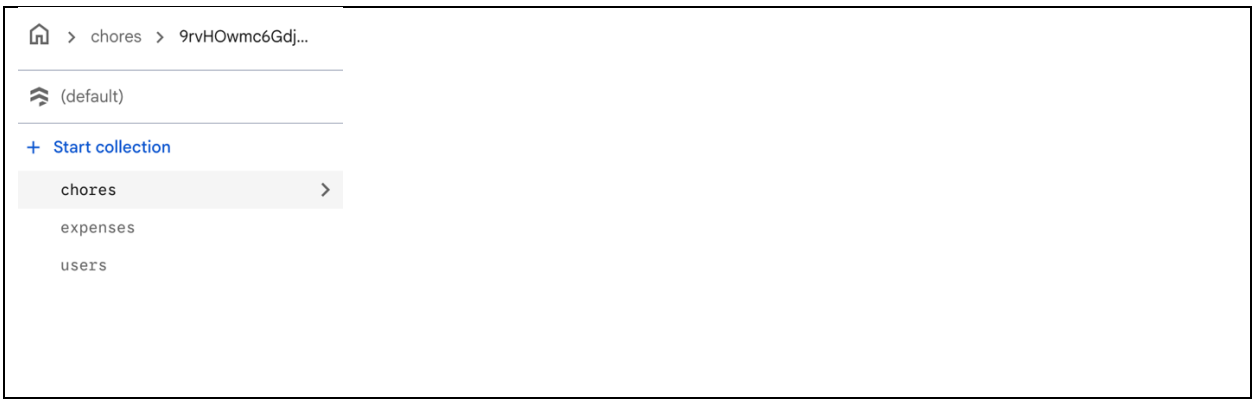
We think a peer evaluation aspect of the project and final grade would be useful for a project like this. Given that it is a semester long project, with team members collaborating all throughout, it would be nice to offer peer reviews to let the instructor know of various team members' attitudes and work efforts, to ensure fair grading.

Lastly, capture multiple screenshots from your personal Firebase account to illustrate the following points. Attach your screenshots after the corresponding items below.

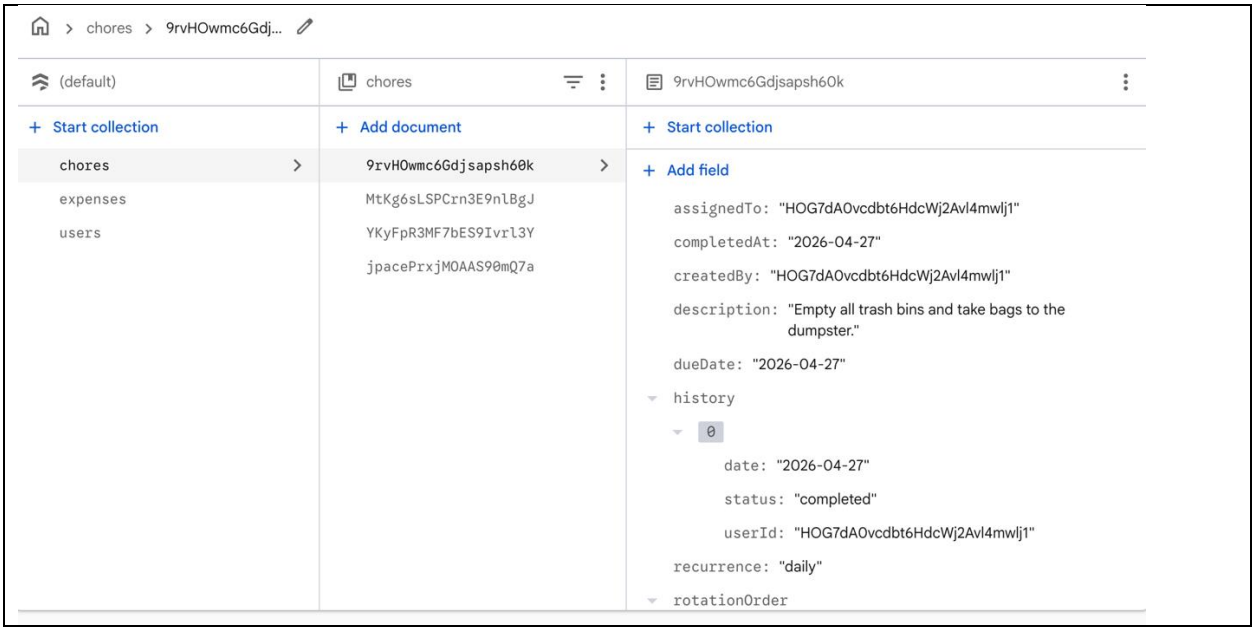
1. Show your existing users.

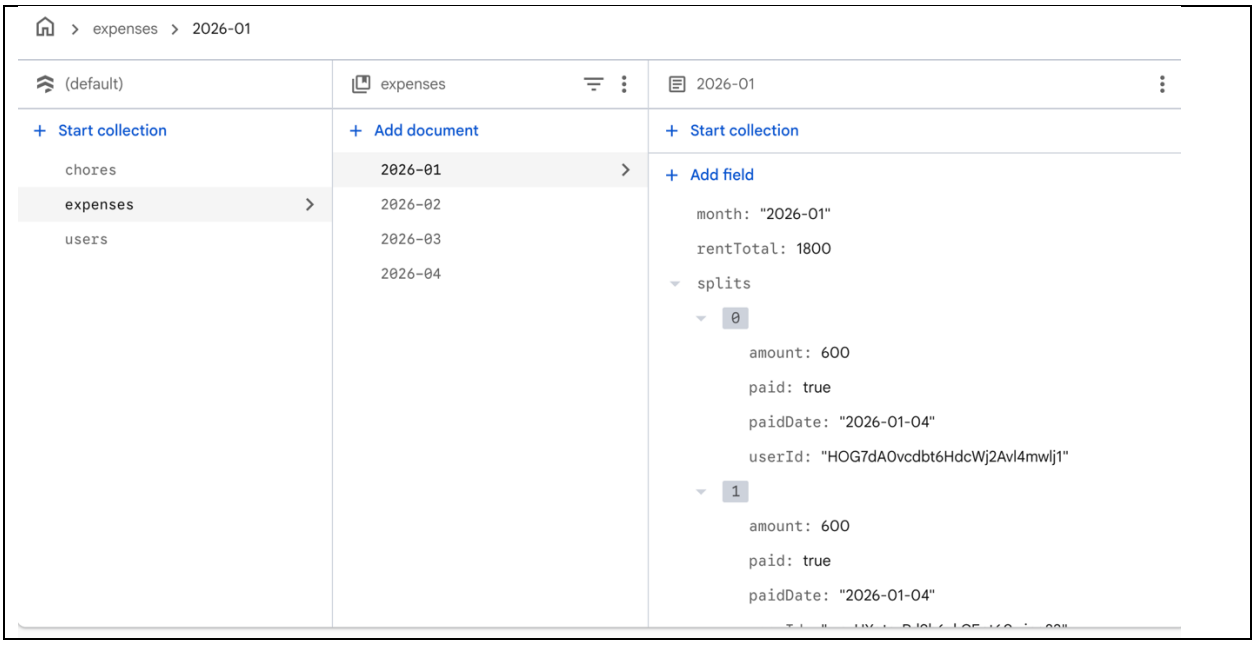


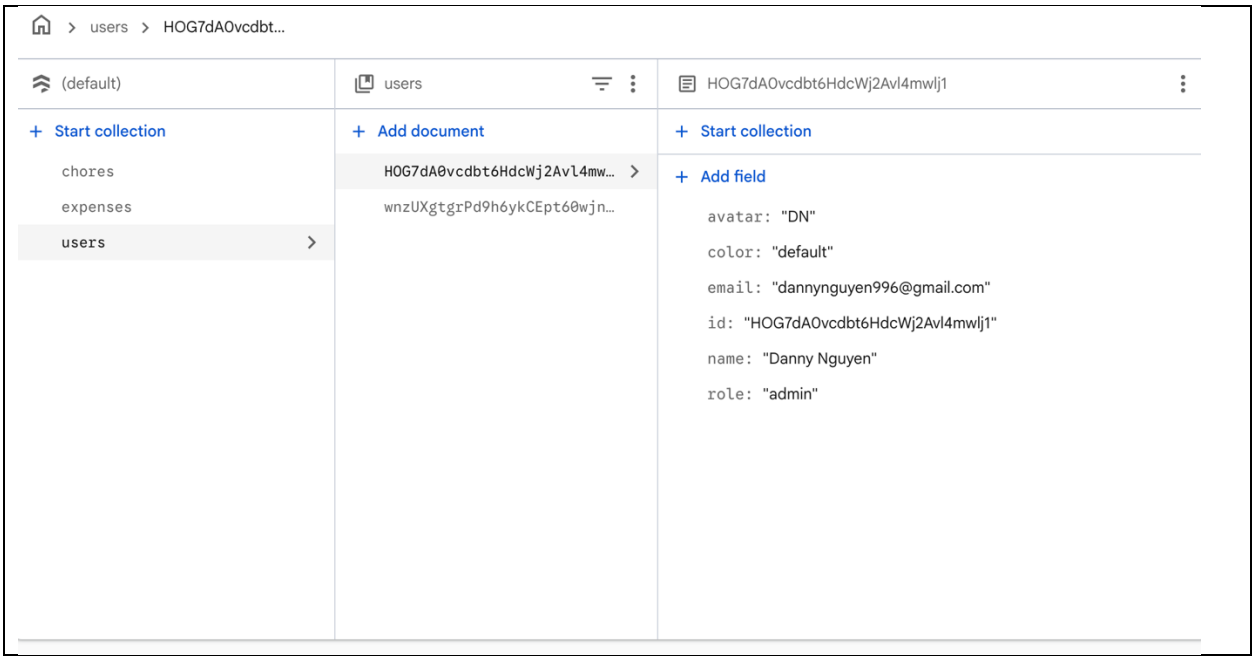
2. Show a comprehensive list of all your collections.



3. Show a sample document from each of your collections. For example, if you have 3 collections, provide 3 corresponding screenshots.







4. Show your security rules configuration.

```
rules_version = '2';

service cloud.firestore {
  match /databases/{database}/documents {

    // This rule allows anyone with your Firestore database reference to view, edit,
    // and delete all data in your Firestore database. It is useful for getting
    // started, but it is configured to expire after 30 days because it
    // leaves your app open to attackers. At that time, all client
    // requests to your Firestore database will be denied.
    //
    // Make sure to write security rules for your app before that time, or else
    // all client requests to your Firestore database will be denied until you Update
    // your rules

    match /expenses/{expenseId} {
      // Anyone can read expenses
      allow read: if true;

      // Only admins can write
      allow write: if
        request.auth != null &&
        get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == "admin";
    }

  }
}

```

```
rules_version = '2';

service cloud.firestore {
  match /databases/{database}/documents {

    // This rule allows anyone with your Firestore database reference to view, edit,
    // and delete all data in your Firestore database. It is useful for getting
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    // leaves your app open to attackers. At that time, all client
    // requests to your Firestore database will be denied.
    //
    // Make sure to write security rules for your app before that time, or else
    // all client requests to your Firestore database will be denied until you Updat
    // your rules
    match /{document=**} {
      allow read, write: if request.time < timestamp.date(2026, 5, 8);
    }
  }
}
```